Mathematical Methods for Finance

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Module 3 - Lesson 11

Convergence, limit theorems and characteristic functions

What happens when the number of observations grows

Learning outcomes for Lesson 11

By the end of the lesson students should be able to

1. distinguish convergence almost surely, in probability, in mean square and in distribution;
2. apply Markov inequality, law of large numbers and central limit theorem;
3. use characteristic functions in basic computations;
4. understand weak convergence and Levy continuity theorem;
5. decide which theorem is appropriate in an exam problem.

How today is organised

Block 1

Sequences of random variables and modes of convergence.

Block 2

LLN, CLT, characteristic functions and weak convergence.

Block 3

Exam-style recognition and computations.

Block 1

Sequences and modes of convergence

Why convergence matters

Intuition: Repeated observation

Many statistical quantities are sequences:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S_n = \sum_{i=1}^{n} X_i.$$

The question is: do these random quantities stabilize as n grows?

Key warning

There are several notions of convergence, and they are not interchangeable.

Almost sure convergence

Definition: Almost sure convergence

$$X_n \to X \quad \text{a.s.}$$

if

$$\mathbb{P}\big(\{\omega : X_n(\omega) \to X(\omega)\}\big) = 1.$$

Intuition: Pathwise meaning

For almost every outcome, the numerical sequence $X_n(\omega)$ converges.

Convergence in probability

Definition: In probability

$$X_n \to X \quad \text{in probability}$$

if for every $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) \to 0.$$

Intuition: Meaning

Large deviations from the limit become unlikely.

Mean-square convergence

Definition: Mean-square convergence

$$X_n \to X \quad \text{in } L^2$$

if

$$\mathbb{E}[(X_n - X)^2] \to 0.$$

Theorem: Implication

$$X_n \to X \text{ in } L^2 \quad \Longrightarrow \quad X_n \to X \text{ in probability.}$$

Convergence in distribution

Definition: Weak convergence

$$X_n \Rightarrow X$$

if

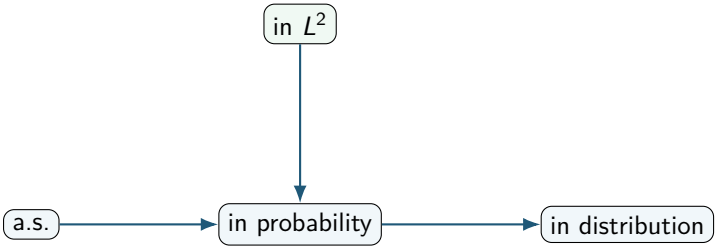
$$F_{X_n}(x) \to F_X(x)$$

at every continuity point of F_X.

Intuition: Distributional meaning

The laws become close, but the random variables need not be close outcome by outcome.

Map of implications



Main message

Stronger convergence gives weaker convergence, but the converse generally fails.

Counterexample: distribution is weaker

Counterexample: Convergence in distribution does not imply convergence in probability

Let $X \sim N(0,1)$ and define $X_n = -X$ for all n. Then X_n has the same distribution as X, so $X_n \Rightarrow X$. But $X_n - X = -2X$, which does not converge to 0 in probability.

Markov inequality

Theorem: Markov inequality

If $Y \geq 0$ and $a > 0$, then

$$\mathbb{P}(Y \geq a) \leq \frac{\mathbb{E}[Y]}{a}.$$

Example: With squares

Apply it to $Y = (X - \mu)^2$ to get Chebyshev-type bounds.

Worked Markov inequality

Exercise: Problem ⋆⋆

Let $Y \geq 0$ with $\mathbb{E}[Y] = 3$. Bound $\mathbb{P}(Y \geq 10)$.

Worked example: Solution

By Markov inequality,

$$\mathbb{P}(Y \geq 10) \leq \frac{\mathbb{E}[Y]}{10} = 0.3.$$

Chebyshev inequality

Theorem: Chebyshev inequality

If $\mathbb{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2$, then

$$\mathbb{P}(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}.$$

Exercise: Quick use $\star\star$

If $\mathrm{Var}(X) = 4$, bound $\mathbb{P}(|X - \mathbb{E}[X]| \geq 5)$.

Worked example: Answer

The bound is $4/25 = 0.16$.

Convergence in probability via variance

Theorem: Useful criterion

If $\mathbb{E}[X_n] \to c$ and $\mathrm{Var}(X_n) \to 0$, then

$$X_n \to c \quad \text{in probability.}$$

Proof idea: Why

Use Chebyshev:

$$\mathbb{P}(|X_n - c| > \varepsilon) \leq \frac{\mathbb{E}[(X_n - c)^2]}{\varepsilon^2} \longrightarrow 0.$$

Block 1 mini-workshop

Exercise: Recognize the convergence

For each statement, identify the convergence type:

1. $\mathbb{P}(|X_n - X| > 0.01) \to 0$.
2. $\mathbb{E}[(X_n - X)^2] \to 0$.
3. $F_{X_n}(x) \to F_X(x)$ at continuity points.
4. $X_n(\omega) \to X(\omega)$ for almost every ω.

Block 2

LLN, CLT and characteristic functions

Law of large numbers

Theorem: Weak law of large numbers

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and finite variance. Then

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i \to \mu$$

in probability.

Intuition: Meaning

Sample averages stabilize around the theoretical mean.

Proof idea of LLN

Proof idea: Variance shrinking

For i.i.d. variables with variance σ^2,

$$\mathbb{E}[\bar{X}_n] = \mu, \qquad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

Then Chebyshev gives

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \to 0.$$

LLN exercise

Exercise: Problem ⋆⋆

Let X_i be i.i.d. with $\mathbb{E}[X_i] = 5$, $\mathrm{Var}(X_i) = 9$. Compute $\mathbb{E}[\bar{X}_n]$, $\mathrm{Var}(\bar{X}_n)$, and bound $\mathbb{P}(|\bar{X}_n - 5| > 1)$.

Worked example: Solution

$$\mathbb{E}[\bar{X}_n] = 5, \qquad \mathrm{Var}(\bar{X}_n) = \frac{9}{n}.$$

By Chebyshev, $\mathbb{P}(|\bar{X}_n - 5| > 1) \le 9/n$.

Central limit theorem

Theorem: CLT

Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and variance $\sigma^2 > 0$. Then

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \Rightarrow N(0,1).$$

Intuition: Meaning

After centering and scaling, the distribution of the sample mean is approximately normal for large n.

CLT standardization

Exercise: Problem ⋆⋆

Let X_i be i.i.d. with $\mu = 10$, $\sigma = 2$. Approximate

$$\mathbb{P}(\bar{X}_{100} \leq 10.4).$$

Worked example: Solution

$$Z = \frac{\sqrt{100}(\bar{X}_{100} - 10)}{2}.$$

The threshold 10.4 gives

$$\frac{10(10.4 - 10)}{2} = 2.$$

So the probability is approximately $\mathbb{P}(Z \leq 2)$.

LLN versus CLT

Theorem	Object	Output
LLN	$\bar{X}_n$	convergence to μ
CLT	$\sqrt{n}(\bar{X}_n - \mu)/\sigma$	normal approximation

Counterexample: Common confusion

The LLN says where the average goes. The CLT describes the fluctuations around that limit.

Weak convergence

Definition: Weak convergence

$X_n \Rightarrow X$ means convergence of distributions. Equivalently,

$$\mathbb{E}[g(X_n)] \to \mathbb{E}[g(X)]$$

for every bounded continuous function g.

Intuition: Testing by functions

Instead of checking all intervals, test the laws through continuous functions.

Characteristic functions

Definition: Characteristic function

The characteristic function of X is

$$\varphi_X(t) = \mathbb{E}[e^{itX}], \qquad t \in \mathbb{R}.$$

Why useful

- It always exists because $|e^{itX}| = 1$.
- It characterizes the distribution.
- It turns sums of independent variables into products.

Characteristic function of a Bernoulli

Exercise: Problem $\star\star$

Let $X \sim \text{Bernoulli}(p)$. Compute $\varphi_X(t)$.

Worked example: Solution

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = e^{it\cdot 0}(1-p) + e^{it\cdot 1}p = 1 - p + pe^{it}.$$

Sums and independence

Theorem: Product rule

If X and Y are independent, then

$$\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t).$$

Exercise: Quick use $\star\star$

If $X_1, \ldots, X_n$ are i.i.d. Bernoulli(p), write the characteristic function of $S_n = \sum_i X_i$.

Worked example: Answer

$$\varphi_{S_n}(t) = (1 - p + pe^{it})^n.$$

Inversion theorem: message

Theorem: Inversion principle

The characteristic function determines the distribution. Under suitable regularity, the density or probabilities can be recovered from φ_X.

Intuition: Message

The characteristic function is not only a summary. It contains the full distributional information.

Levy continuity theorem

Theorem: Levy continuity theorem

If $\varphi_{X_n}(t) \to \varphi(t)$ for every t, and φ is continuous at 0, then φ is the characteristic function of some X and

$$X_n \Rightarrow X.$$

Intuition: Why important

It is often easier to prove convergence of characteristic functions than convergence of distribution functions.

Necessary and sufficient weak convergence: usable form

Exam-level criterion

To prove or recognize weak convergence, common routes are:

1. convergence of distribution functions at continuity points;
2. convergence of expectations of bounded continuous test functions;
3. convergence of characteristic functions plus Levy theorem.

Exercise: Recognition $\star$

Which route is used in the classical proof of the CLT?

Block 2 mini-workshop

Exercise: Four short questions

1. Which theorem says $\bar{X}_n \to \mu$?
2. Which theorem gives a normal approximation for $\bar{X}_n$?
3. Why does a characteristic function always exist?
4. What does Levy theorem connect?

Block 3

Exam-style exercises

Guided exercise 1: convergence types

Exercise: Problem $\star\star$

Let X_n satisfy $\mathbb{E}[(X_n - 3)^2] = 1/n$. Prove that $X_n \to 3$ in probability.

Guided exercise 1: convergence types

Exercise: Problem ⋆⋆

Let X_n satisfy $\mathbb{E}[(X_n - 3)^2] = 1/n$. Prove that $X_n \to 3$ in probability.

Worked example: Solution

For every $\varepsilon > 0$, Markov/Chebyshev gives

$$\mathbb{P}(|X_n - 3| > \varepsilon) \le \frac{\mathbb{E}[(X_n - 3)^2]}{\varepsilon^2} = \frac{1}{n\varepsilon^2} \to 0.$$

Thus $X_n \to 3$ in probability.

Guided exercise 2: LLN bound

Exercise: Problem ⋆⋆

Let X_i be i.i.d. with $\mathbb{E}[X_i] = 2$, $\mathrm{Var}(X_i) = 16$. Find n such that Chebyshev guarantees

$$\mathbb{P}(|\bar{X}_n - 2| > 0.5) \leq 0.04.$$

Solution: guided exercise 2

Worked example: Solution

$$\mathrm{Var}(\bar{X}_n) = \frac{16}{n}.$$

By Chebyshev,

$$\mathbb{P}(|\bar{X}_n - 2| > 0.5) \leq \frac{16/n}{0.5^2} = \frac{16/n}{0.25} = \frac{64}{n}.$$

We need $64/n \leq 0.04$, hence $n \geq 1600$.

Guided exercise 3: CLT probability

Exercise: Problem $\star\star\star$

Let X_i be i.i.d. with mean 50 and standard deviation 10. Approximate

$$\mathbb{P}(\bar{X}_{64} \leq 52.5).$$

Guided exercise 3: CLT probability

Exercise: Problem $\star\star\star$

Let X_i be i.i.d. with mean 50 and standard deviation 10. Approximate

$$\mathbb{P}(\bar{X}_{64} \leq 52.5).$$

Worked example: Solution

$$Z = \frac{\sqrt{64}(\bar{X}_{64} - 50)}{10}.$$

The threshold gives

$$\frac{8(52.5 - 50)}{10} = 2.$$

So the approximation is $\mathbb{P}(Z \leq 2)$.

Guided exercise 4: characteristic functions

Exercise: Problem $\star\star\star$

Let X take values -1 and 1 with probabilities $1/2$ and $1/2$. Compute $\varphi_X(t)$. Then write the characteristic function of $S_n = X_1 + \cdots + X_n$ for i.i.d. copies.

Solution: guided exercise 4

Worked example: Solution

$$\varphi_X(t) = \frac{1}{2}e^{-it} + \frac{1}{2}e^{it} = \cos t.$$

If $X_1, \ldots, X_n$ are independent copies, then

$$\varphi_{S_n}(t) = (\cos t)^n.$$

Independent exercise

Exercise: Try alone ⋆⋆⋆

Let X_i be i.i.d. with $\mathbb{E}[X_i] = 1$, $\mathrm{Var}(X_i) = 4$. Define $\bar{X}_n = n^{-1} \sum_i X_i$.

1. State the LLN conclusion.
2. Bound $\mathbb{P}(|\bar{X}_n - 1| > 0.2)$ with Chebyshev.
3. Use the CLT to approximate $\mathbb{P}(\bar{X}_{100} \leq 1.4)$.

Solution: independent exercise

Worked example: Solution

LLN: $\bar{X}_n \to 1$ in probability. Also

$$\mathrm{Var}(\bar{X}_n) = \frac{4}{n}.$$

Chebyshev gives

$$\mathbb{P}(|\bar{X}_n - 1| > 0.2) \le \frac{4/n}{0.04} = \frac{100}{n}.$$

For $n = 100$, standard deviation of $\bar{X}_{100}$ is $2/10 = 0.2$. The threshold 1.4 gives $z = (1.4 - 1)/0.2 = 2$, so probability $\approx \mathbb{P}(Z \le 2)$.

Exercise: theorem selection

Exercise: Choose the tool $\star\star$

For each question, choose LLN, CLT, Markov/Chebyshev or characteristic functions.

1. Prove that sample averages stabilize.
2. Approximate the distribution of a standardized average.
3. Bound a tail probability using only mean/variance.
4. Prove convergence in distribution using transformed laws.

Solution: theorem selection

Worked example: Answer

1. LLN.
2. CLT.
3. Markov or Chebyshev.
4. Characteristic functions and Levy continuity theorem.

Exam habit

Before computing, name the theorem and check its hypotheses.

Common mistakes

Counterexample: Avoid these

- Saying $X_n \Rightarrow X$ means X_n is close to X outcome by outcome.
- Applying CLT without centering and scaling.
- Using LLN to approximate a probability distribution.
- Forgetting that characteristic functions always exist but moment generating functions may not.

Exit ticket

Exercise: Three questions

1. Give one implication among convergence modes.
2. State the LLN in one sentence.
3. State the CLT in one sentence.

End-of-lesson checklist

You should now be able to

- distinguish the four main convergence modes;
- use Markov and Chebyshev inequalities;
- apply LLN and CLT to sample means;
- compute simple characteristic functions;
- state why characteristic functions are useful for weak convergence.

Suggested practice after Lesson 11

Core practice

From the Module 3 exercise pack: convergence of random variables, LLN, CLT, characteristic functions and conditional expectation review.

Exercise: Minimum training set

- two convergence-mode identification exercises;
- two Chebyshev/LLN exercises;
- two CLT standardization exercises;
- two characteristic-function exercises.

Bridge to Module 4

Next step

Module 4 moves from a single random variable or vector to a whole family

$$(X_t)_{t \in T}.$$

The new objects will be paths, filtrations, martingales, Markov processes, Brownian motion and Poisson processes.

Intuition: Continuity of the course

Probability gives the language. Stochastic processes add time and information.